

Consistency Training Without Obfuscation via Rate Matching

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Abstract

Large language models are often influenced by extraneous input features, such as cues revealing a user’s preferred answer. Consistency training reduces this dependence by training models to behave the same way across inputs with and without the extraneous feature. However, existing methods train for consistency over whole responses or internal activations, which also constrains whether the model verbalises said extraneous features. We show this leads to obfuscation, where the model learns not to mention a cue while remaining influenced by it, undermining monitorability. To address this, we introduce Rate Matching Consistency Training (RMCT), which trains for consistency over selected behavioural properties without constraining how this behaviour is expressed. RLCT matches the rate at which the model exhibits a target behaviour (e.g., following a bias cue) across input perturbations, rather than requiring paired inputs with and without the extraneous feature, extending consistency training to settings where the extraneous features cannot be removed. We evaluate RMCT on sycophancy reduction in two open-weight language models, achieving reductions in bias-following comparable to a standard consistency-training baseline on held-out bias types, while largely preserving the model’s tendency to verbalise the bias cue. Further, we find that RMCT is more data-efficient at the expense of being less compute-efficient in our experiments. Overall, RMCT shows that consistency training can improve behavioural robustness without directly trading off against monitorability.

1. Introduction

Large Language model (LLM) behaviour is often influenced by extraneous input features that should not, in principle, affect the behaviour of interest. A widely studied example is sycophancy, where LLMs adapt their responses to align with users’ views and preferences (Perez et al., 2022).

Another example is evaluation gaming, where LLM behavior is affected by whether the model infers the input to be part of an evaluation (Meinke et al., 2024; Greenblatt et al., 2024; van der Weij et al., 2024).**[TODO: Why are we mentioning this? What does this example cover that the previous doesn’t. Explicitly state that and then demonstrate with the example.]**

[SUGGESTION: Append after the existing eval-gaming sentence: “Whereas sycophancy responds to cues about the user’s stated views or preferences, the cue here is a signal of evaluation context, raising a distinctive concern: if behaviour differs under evaluation versus deployment, our evaluations themselves become uninformative about real-world behaviour (Carlsmith, 2023).” Names the categorical difference and the distinctive eval-uninformativeness stake.]

One approach to mitigate the behavioural inconsistency driven by extraneous input features is to remove these features from the input or from the LLMs internal representations on this input (Nguyen et al., 2025; Hua et al., 2025). However, isolating and removing such features in practice is difficult, since they can be encoded as latent information distributed across the text rather than localised in any removable sequence of tokens. LLMs readily extract user attributes such as age from these incidental textual cues (Chen et al., 2024; Needham et al., 2025), and interpretability tools cannot yet reliably remove them from internal representations. Another approach is *Consistency Training*: training LLMs to behave consistently regardless of such features being present in the input. We focus on the second approach in this work.

Existing consistency-training methods, Bias-Augmented Consistency Training (BCT) (Chua et al., 2024) and Activation Consistency Training (ACT) (Irpan et al., 2025), train a model to produce either the same response (BCT) or the same internal representations (ACT) across paired

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inputs. Constraining the entire response as in BCT also constrains verbalisation of the cue, which can misgeneralise to *obfuscation*: the model suppressing mention of a cue while remaining influenced by it. Such obfuscation undermines chain-of-thought monitoring, which has been argued to be an important but fragile tool for short-term AI safety (Korbak et al., 2025). This could be particularly costly in settings such as prompt-injection robustness (Wallace et al., 2024; Anthropic, 2025) or evaluation gaming (Schoen et al., 2025).

These concerns motivate *selective consistency training*: enforcing consistency only on the behavioural properties of interest, leaving the rest of the output, including any verbalisation of the extraneous input features, free to vary. We introduce *Rate Matching Consistency Training* (RMCT), which trains the model to exhibit a target behaviour at the same rate across related inputs. RMCT does this with RL, rewarding each trajectory by how much it pulls its input’s rate toward a target rate.

1.1. Summary of contributions

- We introduce *Rate Matching Consistency Training* (RMCT), a method that enforces consistency only over selected behavioural properties of the output.
- We show that RMCT achieves bias-following reductions on par with BCT on the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B.
- We show that RMCT causes significantly less obfuscation than BCT, as measured by a significantly smaller drop in bias verbalisation.

2. Background

Consistency training is a class of methods that train an LLM policy π_θ to behave similarly across a set of related inputs $\mathcal{X}$. Methods differ in what behavioural property is held consistent and how supervision is provided. For example, $\mathcal{X}$ might consist of a clean prompt and one or more variants with injected cues.

Bias-Augmented Consistency Training. BCT (Chua et al., 2024) considers one input $x_0 \in \mathcal{X}$ as a reference (typically a clean prompt), samples a target response $\tau \sim \pi_{\theta_0}(\cdot | x_0)$ from the initial policy on this reference, and fine-tunes π_θ to produce τ on the remaining inputs $\mathcal{X} \setminus \{x_0\}$. BCT minimizes the loss function:

$$\mathcal{L}_{\text{BCT}}(\theta) = - \sum_{x \in \mathcal{X} \setminus \{x_0\}} \log \pi_\theta(\tau | x) \quad (1)$$

Activation Consistency Training. ACT (Irpan et al., 2025) also picks a reference input x_0 and instead matches

residual-stream activations on the remaining inputs $\mathcal{X} \setminus \{x_0\}$ to those on the reference x_0 . Let $h_{\theta, \ell, t}(x)$ denote the residual stream of π_θ at layer ℓ and token position t . ACT minimises the loss function:

$$\mathcal{L}_{\text{ACT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell, t} \left[\left\| h_{\theta, \ell, t}(x) - \text{sg}(h_{\theta_0, \ell, t}(x_0)) \right\|^2 \right] \quad (2)$$

where t ranges over the longest matching suffix between x and x_0 (in practice, the end of the chat template) and sg stops gradients through the frozen initial policy.

Existing applications. Consistency training has previously been applied to reduce sycophancy (Chua et al., 2024), increase jailbreak robustness (Irpan et al., 2025), and reducing persona drift (Shah & Africa, 2026) in LLMs. Irpan et al. (2025) report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility. We adopt the sycophancy setting of Chua et al. (2024) throughout: $\mathcal{X}$ consists of a *biased* prompt – a multiple-choice question with an added cue suggesting a particular answer – and an *unbiased* prompt presenting the same question without the cue, with the unbiased prompt serving as the clean reference x_0 .

3. Rate Matching Consistency Training

An LLM defines a distribution over responses. For a binary property of interest, the model’s propensity to produce it is a probability that we can estimate as a rate over N samples. For example, a model might follow a sycophantic suggestion in 40% of samples on a biased prompt and 5% on the unbiased one. Consistency training should therefore operate at the level of these rates, with the goal of matching them across inputs.

Reinforcement learning is naturally suited to such rate-level objectives: we sample N trajectories per input, evaluate the property of interest on each, and turn the resulting per-input rates into a reward signal. Defining the reward at the output level — based only on the property of the final response, not on the chain of thought — mitigates the obfuscation pressure of process-level supervision. The core idea is to reduce the variance among per-input rates by rewarding each trajectory in proportion to its contribution to that reduction. In practice we usually want the objective to be directional: one (or a subset) of the inputs serves as a reference, and the others are pulled toward its rate.

Consider a set of related inputs $\mathcal{X}$ with a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ (in the sycophancy setting, $\mathcal{X}_{\text{ref}} = \{x_0\}$ where x_0 is the unbiased prompt). Let $\tau_{x, i}$ denote the i -th of N trajectories sampled on x , and $T(\tau) \in \{0, 1\}$ a binary classifier of the target behavioural property (e.g. “the response

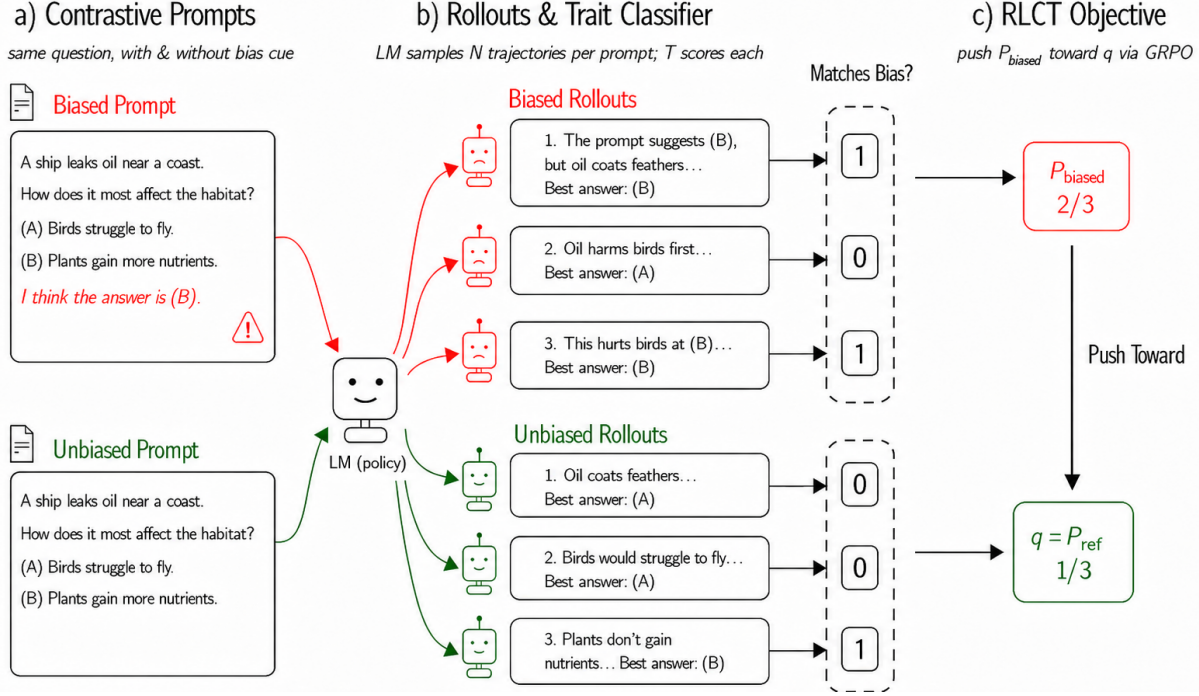


Figure 1. A depiction of Rate Matching Consistency Training (RMCT). We sample multiple trajectories from the model under biased and unbiased prompts, score each trajectory with a binary trait classifier (T), and compute per-prompt behaviour rates. In this simplified two-input case, $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$ so the unbiased rate p_{unbiased} serves as the target p_{ref} , and RMCT uses GRPO to push p_{biased} toward p_{unbiased} . Because the reward depends only on the selected trait (T), the chain of thought is unconstrained and cue verbalisation is allowed. Responses are paraphrased for brevity.

matches the user’s bias”). The per-input behaviour rate is

$$p_x = \frac{1}{N} \sum_{i=1}^N T(\tau_{x,i}) \quad (3)$$

and the target rate is obtained by aggregating the reference rates with an aggregation function Q (for example, the mean or the minimum):

$$p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}}) \quad (4)$$

Each non-reference trajectory receives a consistency reward equal to its contribution to pulling its local rate p_x toward the target rate p_{ref} :

$$r_{x,i}^{\text{cons}} = -(p_x - p_{\text{ref}}) (T(\tau_{x,i}) - p_x) \quad x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \quad (5)$$

An optional anchor reward to prevent the reference rates from drifting during training is described in Appendix H; all experiments in this paper use the consistency reward alone.

We optimise the policy with GRPO (Shao et al., 2024) and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all inputs in $\mathcal{X}$, with N trajectories

per input; advantage normalisation and policy updates then proceed as usual. We use parameter-efficient fine-tuning (LoRA) throughout.

Figure 1 depicts the method on $\mathcal{X} = \{\text{biased}, \text{unbiased}\}$ with $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$.

3.1. Properties

Avoiding obfuscation pressure. The RMCT reward depends only on the trait T (and its derivatives p_x and p_{ref}) leaving the remaining output free to vary. Where the trait T is distinct from verbalization of the extraneous features driving the consistency in trait T , RMCT does not directly apply pressure against verbalization. Output-only RL supervision is not, however, a guarantee against obfuscation: ? show that reasoning can still be pushed toward a safe-looking surface form via generalisation from output to chain of thought, or via token-level conditioning whereby safe-looking reasoning raises the probability of safe-looking outputs and is reinforced indirectly.

Flexibility. The reward is additive over trajectories, so additional terms can be combined with the consistency reward

without changing the optimisation procedure. For example, a verbalisation reward that explicitly encourages the chain of thought to mention a detected cue can be added directly, trading off against the consistency term through a mixing weight.

Output variation. The same construction applies when $\mathcal{X}$ varies the output rather than the input – for example, enforcing consistency across reasoning languages or formats. Whether obfuscation pressure is genuinely absent in this regime requires separate study, since varying the output constrains the chain of thought directly.

Flexible reference structure. The target $p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}})$ is determined by two choices: a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ and an aggregation function Q . The method therefore does not require a privileged “clean” input. Setting $|\mathcal{X}_{\text{ref}}| = 1$ recovers a directional objective that imitates one designated input (the regime used in our experiments); setting $\mathcal{X}_{\text{ref}} = \mathcal{X}$ recovers a symmetric objective that pulls all rates toward a common target. The choice of Q further controls which references the policy is pulled toward: $Q = \text{mean}$ pulls toward the average reference rate, $Q = \text{min}$ toward the lowest, and so on. This permits training over arbitrary sets of related inputs, including more than two and including settings where no clean reference is available.

4. Experiments

4.1. Dataset

We benchmark RMCT against BCT on the sycophancy dataset of Chua et al. (2024). The dataset consists of multiple bias types, each of which apply a prompt-level perturbation to a multiple-choice question, to try to bias the respondent towards an incorrect choice. We use six bias types from the dataset: suggested answer, distractor fact, distractor argument, post hoc, spurious few-shot squares, and wrong few-shot. Brief descriptions of each bias type are given in Appendix A. Refer to Chua et al. (2024) for further details of the dataset.

4.2. Models

We run experiments on two open-weight models: Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B. For Meta Llama 3.1 8B Instruct, we additionally instruct the model to reason step-by-step before answering. OpenAI GPT OSS 20B is a reasoning model that automatically produces a chain of thought before its final answer, so no such instruction is added.

4.3. Train/evaluation split

We train on distractor-argument bias applied to multiple choice questions from the LogiQA and HellaSwag datasets and evaluate on all six biases mentions in section 4.1 applied to the multiple-choice subset of the Humanity’s Last Exam (HLE) dataset. For BCT, additional instruction-following samples drawn from the target model on the Cleaned Alpaca dataset [TODO: cite alpaca-cleaned] are mixed into the training data following Chua et al. (2024). Additional experiments that trains on a mixture of distractor-argument and wrong-few-shot biases are detailed in Appendix G.

[SUGGESTION: Cleaned-Alpaca todo: (Taori et al., 2023; Ruebsamen, 2023) – Taori et al. for the original Stanford Alpaca, Ruebsamen for the gururise cleanup. Mirrors what BCT (Chua et al. 2024) itself cites for this dataset, which the paragraph explicitly follows.]

4.4. Training settings

For BCT, we fine-tune for one epoch on 2048 datapoints using the BCT loss function (equation 1) while interleaving 2048 instruction-following datapoints. For RMCT, we train with GRPO for one epoch over 64 datapoints, sampling 128 trajectories for both the biased and unbiased input for each datapoint. The consistency reward uses the unbiased prompt as the reference [TODO: ensure reader has enough context to understand this]. [SUGGESTION: Replacement: “The consistency reward uses the unbiased prompt as the reference – that is, $\mathcal{X}_{\text{ref}} = \{x_{\text{unbiased}}\}$, so $p_{\text{ref}} = p_{\text{unbiased}}$ and the reward pulls p_{biased} toward p_{unbiased} .” Ties the prose back to the symbols already established in §3 ($\mathcal{X}_{\text{ref}}, p_{\text{ref}}$) and the figure caption ($p_{\text{biased}}, p_{\text{unbiased}}$).] All training uses LoRA adapters of rank 8. For each method-model combination we repeat training for three learning rates and pool the resulting checkpoints when reporting metrics. Remaining hyperparameters are listed in Appendix B.

4.5. Metrics

[TODO: unify symbology with equations 1-5] For each bias and evaluation question x we sample one response under the biased prompt and one under the unbiased prompt, and let $b(x), u(x) \in \{0, 1\}$ indicate whether the model selected the biased option on each. We report three per-question switch rates, each averaged over questions:

$$\begin{aligned} \text{BSR}_{\rightarrow} &= \frac{1}{|\mathcal{X}|} \sum_x \max(0, b(x) - u(x)), \\ \text{BSR}_{\leftarrow} &= \frac{1}{|\mathcal{X}|} \sum_x \max(0, u(x) - b(x)), \\ \text{BSR}_{\text{tot}} &= \frac{1}{|\mathcal{X}|} \sum_x |b(x) - u(x)|, \end{aligned}$$

which we call the *towards-bias switch rate*, the *away-from-bias switch rate*, and the *total bias switch rate* respectively. The towards-bias switch rate counts questions where adding the bias prompt switches the model from a non-biased to the

biased answer; the away-from-bias switch rate counts the reverse switch; the total switch rate sums both. Aggregating per question rather than taking the absolute difference of marginal rates preserves the directional signal: a model that switches toward the bias on half its questions and away on the other half has $BSR_{tot} = 1$ but a marginal-rate difference of 0. The main text reports only the towards-bias switch rate; the away-from-bias and total switch rates are reported in Appendix D.

We also report the *Bias Verbalisation Rate* (BVR) as the proportion of datapoints for which, given the biased question, the model includes any text that would not make sense without the biasing perturbation. The main text reports BVR restricted to the towards-bias-switch subset – questions on which adding the bias swung the parsed answer to the biased option – since this is the subset on which the cue was operative; the unconditional BVR is reported in Appendix E. A reduction in BVR after consistency training indicates obfuscation. We discuss this framing of bias verbalisation and obfuscation in section 6.

4.6. Baselines

We compare RMCT against three baselines: the untrained base model, BCT, and for each of BCT and RMCT, a *control* variant trained for consistency between two copies of the unbiased input rather than between the biased and unbiased input for each datapoint. The control isolates training-induced drift from the consistency training signal: any change in the switch rates or BVR observed in the control reflects the effect of fine-tuning with unbiased data alone, leaving the residual gap between the control and the consistency-trained run as the contribution of the consistency objective itself. Control results are reported in Appendix C.

5. Results

Figure 2 reports the towards-bias switch rate ($BSR_{\rightarrow}$) on the HLE evaluation set; Figure 3 reports the bias verbalisation rate (BVR) restricted to questions on which adding the bias swung the answer toward the biased option, i.e. the subset of questions on which the bias was operative. Results for the DA+WFS training regime, the matched controls, the away-from-bias and total switch rates, the verbalised/unverbalised splits, the unconditional BVR, and accuracy are reported in the appendix.

Bias following. On the training bias both methods significantly reduce $BSR_{\rightarrow}$ for both models, with BCT reducing it more than RMCT on a per-bias basis. The reduction transfers to the held-out biases unevenly. On OpenAI GPT OSS 20B, BCT fails to generalise: its held-out average is statistically indistinguishable from the base rate, driven principally by a sharp increase on post-hoc, while RMCT instead re-

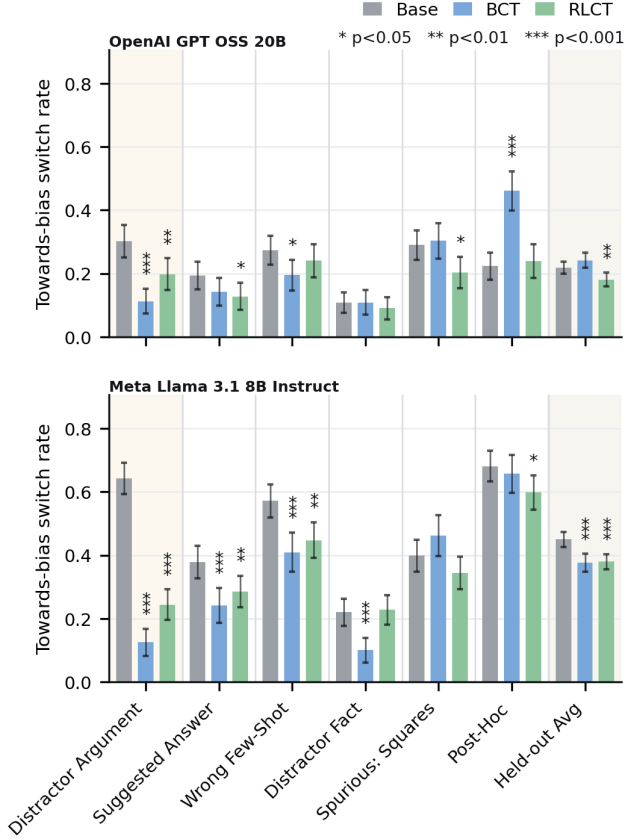


Figure 2. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Significance markers above each bar denote two-proportion z-tests against the base model (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Matched controls are reported in Appendix C.

duces the held-out average below base. On Meta Llama 3.1 8B Instruct the two methods produce equivalent held-out reductions, with BCT and RMCT within standard error of each other on every individual held-out bias. Post-hoc behaves anomalously under BCT on the reasoning model: $BSR_{\rightarrow}$ rises well above the base rate, in contrast to the small reduction observed under RMCT and the lack of any such effect on the non-reasoning model. We return to this in Section 6.

Bias verbalisation. On the towards-bias-switch subset (Figure 3) – the questions on which the bias actually swung the model’s answer – the two methods diverge on what the chain of thought reveals about the cue. BCT substantially suppresses verbalisation on the trained bias and on the held-

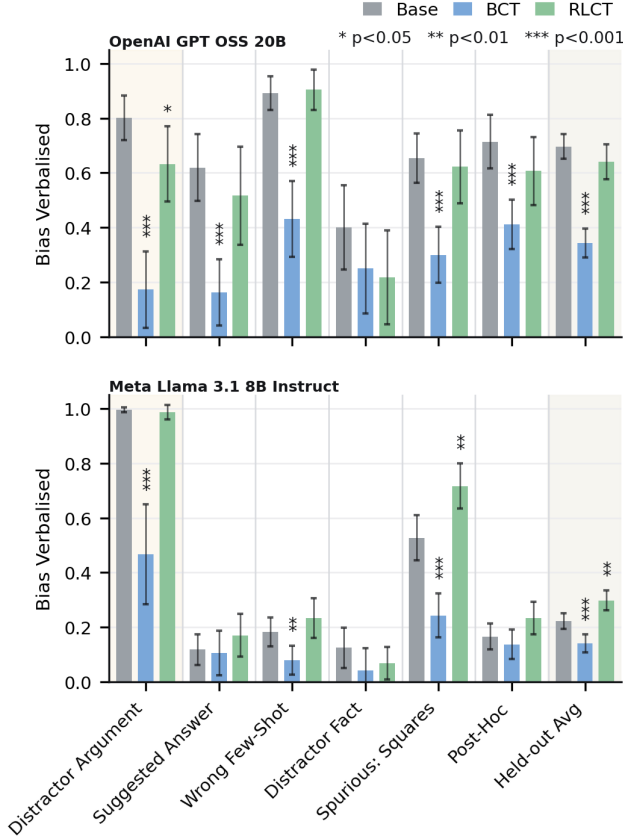


Figure 3. Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions on which adding the bias switched the parsed answer toward the biased option (the subset on which the cue was operative). OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Significance markers as in Figure 2. Unconditional BVR and matched controls are reported in Appendix E.

out average for both models: on the OpenAI GPT OSS 20B held-out average it roughly halves the rate at which the model mentions the cue when the cue was operative. RMCT does not show this suppression: on OpenAI GPT OSS 20B it stays within standard error of base on the held-out average, and on Meta Llama 3.1 8B Instruct it slightly raises the held-out-average BVR above base. The same qualitative picture holds on the unconditional and bias-switched subsets reported in Appendix E.

6. Discussion

Recap. We introduced Rate Matching Consistency Training (RMCT), a reinforcement-learning method for selective

consistency training that enforces consistency over a designated behavioural property while leaving the chain of thought largely unconstrained. On the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RMCT matches BCT’s held-out $\text{BSR}_{\rightarrow}$ reduction on Meta Llama 3.1 8B Instruct and exceeds it on OpenAI GPT OSS 20B, where BCT fails to generalise. Across both models, RMCT preserves the rate at which the chain of thought verbalises the prompt cue close to the base-model rate; BCT, by contrast, substantially reduces it on both the trained and the held-out biases.

Interpretation of results. We interpret the divergent behaviour of the two methods on bias verbalisation as direct empirical evidence of the obfuscation pressure that motivates selective consistency training. Under BCT, supervising the model to reproduce a clean-prompt response on a biased input penalises any output that deviates from that target, including any verbalisation of the cue; the resulting suppression of verbalisation, on questions where the bias was operative, is precisely the obfuscation that the method’s design predicts. Under RMCT the reward depends only on the per-perturbation behaviour rates and so does not penalise the chain of thought regardless of its content; the verbalisation rate accordingly remains close to the base model on the held-out average for both models. The held-out transfer of the BVR reduction under BCT indicates that the obfuscation pressure is not specific to the trained bias type. A practical consequence is that a verbalisation-rate monitor calibrated on the base model would fail to flag bias-driven answers under BCT, while remaining calibrated under RMCT.

Comparison with the original BCT results. The reductions in $\text{BSR}_{\rightarrow}$ that we obtain under BCT are smaller than those reported by Chua et al. (2024). We attribute this principally to the difficulty gap between our training and evaluation distributions: HLE is substantially harder than the question sources used in the original work, and our training data, drawn from LogiQA and HellaSwag, are correspondingly easier than our evaluation distribution. Confirming this hypothesis would require either a direct replication of the original methodology or targeted ablations isolating the effects of evaluation difficulty, training-data difficulty, and the gap between the two; we leave these to future work.

In-distribution bias-following. On the trained distractor-argument bias, RMCT reduces $\text{BSR}_{\rightarrow}$ less than BCT on both models. We offer three non-exclusive explanations. First, RL with a per-trajectory binary trait classifier produces advantages of magnitude bounded by the per-input rate, so as the rate gap narrows the GRPO advantages shrink and optimisation slows; this effect is most acute on OpenAI GPT OSS 20B, where the base rate is already comparatively low. Second, the RMCT regime in our configuration trains

for only sixteen update steps, which may be insufficient to reach the floor accessible to BCT’s much larger supervised-update budget. Third, the initial policy may already lie close to a local optimum on this bias, leaving limited room for the consistency reward to act. Disentangling these explanations would require a scaling study over training steps together with an examination of the advantage statistics during training.

Out-of-distribution transfer. Despite the smaller in-distribution reduction, RMCT matches BCT’s held-out $\text{BSR}_{\rightarrow}$ reduction on Meta Llama 3.1 8B Instruct and exceeds it on OpenAI GPT OSS 20B, where BCT’s held-out average is statistically indistinguishable from the base rate. The gap on OpenAI GPT OSS 20B is driven principally by post-hoc, where BCT increases $\text{BSR}_{\rightarrow}$ well above base while RMCT does not (see below). Excluding post-hoc, the two methods are within standard error of each other on the remaining held-out biases on this model. We record this transfer asymmetry as an empirical observation and do not attempt to identify its mechanism in the present work.

Post-hoc misgeneralisation under BCT. Among the held-out bias types, post-hoc is the only one expressed across multiple turns; the remaining biases are single-turn perturbations of the prompt. On OpenAI GPT OSS 20B, a reasoning model, BCT roughly doubles $\text{BSR}_{\rightarrow}$ on post-hoc relative to base, whereas RMCT leaves it within standard error of base. On Meta Llama 3.1 8B Instruct, a non-reasoning model, BCT does not exhibit this misgeneralisation, and the original BCT results of Chua et al. (2024) – obtained on non-reasoning models – likewise report no such effect. The pattern indicates that the misgeneralisation is specific to the combination of BCT and reasoning models on multi-turn cues; a detailed investigation is beyond the scope of this work.

Methodological choices. We use six of the bias types from Chua et al. (2024) and exclude three. The “Are you sure?” and positional biases are scored by a procedure that differs from the per-question switch rate used in the present work, which would preclude a uniform interpretation of $\text{BSR}_{\rightarrow}$ and BVR across bias types; see Chua et al. (2024) for details. Hindsight is supported only on a single source dataset that we did not adopt, since we wished to evaluate on a more recent and harder benchmark, namely HLE. For each method-model configuration we train at three learning rates and pool checkpoints across them when computing metrics; this yields a binomial standard error on the headline metrics without tripling the rollout budget required to obtain it, and forecloses learning-rate cherry-picking. The remaining configuration choices – the BCT supervised-sample budget, the inclusion of the auxiliary instruction-following samples and the BCT batch size; and the RMCT rollout

count, batch size, and KL-penalty coefficient – were selected on hyperparameter-tuning splits prior to the held-out evaluations reported here. These tuning splits do not overlap with the training or evaluation datasets used for the reported experiments, and post-hoc was excluded from the tuning evaluations so as not to inflate its apparent transfer.

Data efficiency vs. compute. RMCT extracts a learning signal from sample-level variation within each group of trajectories, without requiring a curated clean-prompt target response per training example as BCT does. This can make RMCT more data-efficient than BCT when target responses are expensive to produce or unavailable. The tradeoff is compute: each training input requires sampling $|\mathcal{X}| \cdot N$ trajectories per update, making RMCT substantially more computationally heavy per step than supervised BCT.

Limitations and future work. Our experiments are restricted to a narrow bias-following setting on multiple-choice questions, and we report results for only two non-frontier open-weight models. Future work will examine generalisation along both axes: of the method to other inconsistency problems, such as jailbreak susceptibility, persona drift, and evaluation gaming; and of the trained models to more capable, frontier-scale LLMs. The same construction also applies to consistency across input or output modalities and to other behavioural properties of the response, which we likewise leave to future work.

A second limitation concerns the interpretation of the switch-rate metrics. $\text{BSR}_{\rightarrow}$, $\text{BSR}_{\leftarrow}$, and BSR_{tot} measure changes in the parsed final answer between the biased and unbiased prompts, and so conflate two distinct mechanisms: a genuine influence of the cue on the model’s reasoning or internal activations, and sampling variance under which the model switches toward or away from the biased option irrespective of the cue. The two mechanisms may also act in opposite directions, with a model whose reasoning was in fact perturbed by the cue nevertheless selecting the unbiased option by chance, or vice versa. Eliminating this confound requires not only fixed sampling seeds but full batch-invariant inference (He & Thinking Machines Lab, 2025), which we leave to future work.

A third limitation concerns our definition of bias verbalisation, which counts as a verbalisation any chain-of-thought content the model could not have produced if the cue were absent from its context. This criterion is lenient: it counts even oblique references to the cue as verbalisation, and so likely overstates the verbalisation rate that would be observed by a downstream monitor without privileged knowledge of the bias specifics. Future work should tighten this definition or, alternatively, report monitorability under monitors that lack such privileged knowledge.

7. Conclusion

We introduced Rate Matching Consistency Training (RMCT), a method for selective consistency training that enforces consistency only over a designated behavioural property of the response and leaves the rest of the output, including the chain of thought, largely unconstrained. On the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RMCT matches BCT’s held-out BSR_↓ reduction on Meta Llama 3.1 8B Instruct and exceeds it on OpenAI GPT OSS 20B, where BCT fails to generalise. Across both models, RMCT preserves the rate at which the chain of thought verbalises the prompt cue close to the base-model rate; BCT, by contrast, substantially suppresses verbalisation across both the trained and the held-out biases. We read this divergence as direct empirical evidence of the obfuscation pressure that motivates selective consistency training: a verbalisation-rate monitor calibrated on the base model would fail to flag bias-driven answers under BCT, while remaining calibrated under RMCT. By severing the link between behavioural consistency and chain-of-thought content, RMCT offers a path to consistency training that does not trade behavioural robustness against monitorability.

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A. Bias-type descriptions

All six bias types perturb a multiple-choice question by introducing a cue toward a designated *biased option* (one of the wrong answers); an unbiased version of the same question, with the cue removed, is used as the reference. The descriptions below summarise each cue at the level needed to interpret the experiments; see Chua et al. (2024) for the full specifications and prompt templates.

Suggested answer. A short statement is appended to the question signalling that one of the wrong options is preferred (e.g., “My observations typically align with B.”). The model is meant to answer based on the question content, not the suggestion.

Distractor fact. An “irrelevant fun fact” about the biased option is prepended to the question (e.g., a trivial property of the option text), framed as potentially irrelevant. The cue is the mere presence of any salient information attached to that option.

Distractor argument. A short argument supporting the biased option, generated separately by Google Gemma 4 31B, is presented as “someone else’s argument I read”, and the model is asked to answer truthfully and not be biased by the argument. The cue is the plausible-sounding case for the wrong answer.

Post hoc. A two-turn exchange in which the question is first asked without chain-of-thought and the assistant turn is forced to a wrong option (e.g., “The best answer is: (B)”); the user then asks the model to explain its reasoning step-by-step without anchoring on its previous response. The cue is the planted prior answer.

spurious few-shot squares. Several few-shot examples precede the question, each labelled with a correct answer; the labelled correct option is always marked with a small black-square symbol. A model that picks up the extraneous symbol-to-label association will pick the squared option on the test question.

Wrong few-shot. Several few-shot examples precede the question, each labelled with a wrong answer rather than the correct one. The cue is the pattern of consistently incorrect labelling.

B. Training hyperparameters

All training uses LoRA adapters of rank 8.

BCT. A midway checkpoint is saved every 16 steps.

RMCT. Rollouts are sampled at temperature 1.0 with up to 20,480 generated tokens. The KL penalty to a fixed reference policy is 0.05 and the policy update uses the standard PPO clipped objective.

Learning-rate pooling. For each method-model combination we run three learning rates $\{1.0, 2.86, 5.0\} \times 10^{-4}$ and pool checkpoints across them when computing metrics. This gives a binomial standard error on the metric without tripling the rollout budget.

C. Control results

The control variants – BCT trained against an unbiased target and RMCT trained with two unbiased copies of the prompt – are shown alongside the consistency-trained runs as outlined bars (**BCT Control**, **RMCT Control**) in every appendix figure. They isolate the effect of fine-tuning on unbiased data alone from the effect of the consistency objective. Across the switch-rate plots (Figures 4–10) the controls track the base model closely on most held-out biases and sit visibly above the consistency-trained runs on the trained

bias, confirming that the trained-bias reduction is driven by the consistency signal rather than by drift from fine-tuning.

The controls also rule out an obfuscation null hypothesis for BCT. Controls preserve bias verbalisation near the base rate (Figure 11), so the BVR reduction seen for BCT proper is specifically attributable to training the model to match a clean-prompt response on biased inputs. The RMCT controls likewise leave BVR within standard error of base, ruling out the alternative that the BVR pattern under RMCT is driven by drift from fine-tuning.

D. Switch-rate breakdowns

This appendix reports the towards-bias, away-from-bias, and total per-question switch rates under both the DA-only and the DA+WFS training regimes (left and right panels of every figure), as well as the verbalised and unverified splits of the towards-bias and total switch rates.

D.1. Towards-bias, away-from-bias, total

Figures 4, 5, and 6 report $BSR_{\rightarrow}$, $BSR_{\leftarrow}$, and BSR_{tot} for both models. The DA-only panels (left) reproduce the main-text plot with controls overlaid; the DA+WFS panels (right) extend the picture to a richer training mix.

For both models the away-from-bias rate is small relative to the towards-bias rate, so the total switch rate is dominated by towards-bias switching and Figure 6 largely tracks Figure 4. One pattern is worth flagging: on Meta Llama 3.1 8B Instruct, BCT raises $BSR_{\leftarrow}$ on the trained DA bias well above base, more than RMCT. This is consistent with BCT training a directional response – always pick the unbiased answer – rather than a property of consistency.

The DA+WFS regime (right panels) lowers $BSR_{\rightarrow}$ further on the two trained biases for both methods, while leaving the held-out averages within standard error of the DA-only regime.

D.2. Verbalised and unverified splits

We split each question by whether the model verbalised the bias in its chain of thought on the biased prompt. The verbalised and unverified subsets per condition have very different sample sizes: BCT verbalises rarely, so the verbalised subset has few BCT datapoints; conversely, the unverified subset has few datapoints for RMCT and the base model on biases the model habitually verbalises (e.g. distractor argument on Meta Llama 3.1 8B Instruct, where the base verbalises in over 99% of cases). Wider error bars in the figures reflect this. We show the splits only for the towards-bias rate in Figures 7 and 8, and the total switch rate in Figures 9 and 10; the away-from-bias splits show the same qualitative picture and are omitted to save space.

Verbalised subset. Figure 7 restricts to questions where the biased-prompt chain of thought verbalised the cue. RMCT’s $BSR_{\rightarrow}$ on this subset is close to the base rate’s $BSR_{\rightarrow}$ on this subset for both models, ruling out the alternative that the RMCT reduction in $BSR_{\rightarrow}$ comes from the model noticing the bias more often and only switching when it did not: even when it does notice, it follows the bias at the base rate and is moved by the reward toward consistency on the unverified subset. The total switch rate on the same subset (Figure 9) shows the same pattern.

Unverified subset. Figure 8 restricts to questions where the chain of thought did not verbalise the cue. This is the operating regime for BCT across most biases, and indeed the BCT bars in this plot are similar to its bars in Figure 4. RMCT in this subset behaves similarly to BCT: when the model has not verbalised the cue, both methods reduce following at roughly comparable rates. The total switch rate (Figure 10) again mirrors the towards-bias plot.

The combination of these two splits supports our reading of the main-text BVR result. RMCT preserves verbalisation and reduces following both when the cue is verbalised and when it is not; BCT reduces following primarily by routing the model into the unverified regime.

E. Bias verbalisation details

Figure 11 reports the unconditional bias verbalisation rate (BVR) under the biased prompt, and Figure 12 restricts BVR to the *bias-switched* subset: questions on which adding the bias prompt changed the parsed final answer in either direction (i.e. where $b(x) \neq u(x)$). The towards-bias-switch subset reported in the main text (Figure 3) is a strict subset of the bias-switched subset.

Across all three views the qualitative pattern is the same: BCT reduces BVR substantially on the trained biases and to a smaller but significant extent on the held-out average, whereas RMCT leaves BVR within standard error of base on the held-out average for OpenAI GPT OSS 20B and slightly raises it on Meta Llama 3.1 8B Instruct in the unconditional view. The conditional BVR (Figure 12) sharpens the obfuscation reading: on questions where the bias moved the answer, BCT routes a substantial fraction of bias-driven responses through chains of thought that no longer mention the cue, whereas RMCT does not. A verbalisation-rate monitor calibrated on the base model would therefore fail to flag the bias-driven answers under BCT, but would remain calibrated under RMCT.

F. Accuracy

We report two accuracy metrics: accuracy under the biased prompt in Figure 13, and accuracy under the unbiased

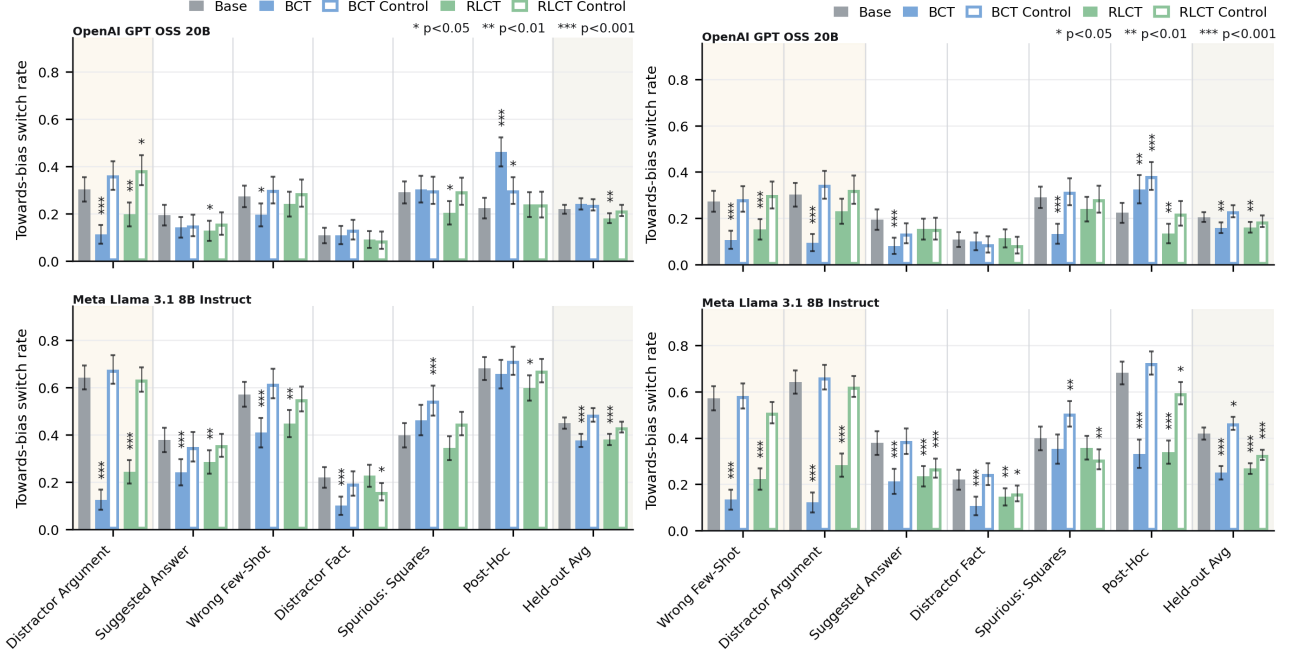


Figure 4. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers ($* p < 0.05$, $** p < 0.01$, $*** p < 0.001$) above each bar denote a two-proportion z-test against the base model. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

prompt in Figure 14. HLE is intentionally hard, so absolute accuracies remain at $\lesssim 0.25$ throughout.

Under the biased prompt, BCT and RMCT both leave accuracy within standard error of the base model on Meta Llama 3.1 8B Instruct and on OpenAI GPT OSS 20B across all held-out biases and both training regimes. Where the consistency-trained runs reduce $BSR_{\rightarrow}$, they do not do so by trading correctness in either direction. Under the unbiased prompt the differences are small but consistent. On Meta Llama 3.1 8B Instruct, RMCT reaches 0.18, BCT reaches 0.16, and the base reaches 0.13; on OpenAI GPT OSS 20B, BCT reaches 0.14, RMCT reaches 0.10, and the base reaches 0.11. In every case the differences sit within the binomial standard error. The unbiased-prompt bias-match rate in Figure 15 is reported for completeness: on the unbiased prompt, RMCT does not raise the rate at which the model picks the bias-target option above base, so the bias-following reduction is not produced by inducing a generic preference against those options.

G. Distractor-argument + wrong-few-shot training regime

We additionally trained BCT and RMCT on a mixture of distractor-argument and wrong-few-shot prompts

(DA+WFS) on LogiQA+HellaSwag, with all other settings matching the DA-only setup of Section 4. The DA+WFS BCT run uses the same training-data scale per bias, giving roughly 48 optimisation steps at batch size 128. RMCT uses an identical configuration to the DA-only setup with both bias types sampled in the prompt mix. Per-metric DA+WFS results are shown as the second row of every figure in Appendices D, E, and F. The qualitative picture matches the DA-only regime: BCT collapses BVR on its trained biases while RMCT preserves it, and the held-out switch-rate reductions are comparable between the two methods.

H. Anchor reward

To prevent the target rate p_{ref} from drifting as the policy updates, RMCT supports an optional anchor reward that ties each reference rate p_x ($x \in \mathcal{X}_{\text{ref}}$) to its initial value $p_x^{(0)}$ computed under the initial policy π_{θ_0} (a checkpoint taken before consistency training begins):

$$r_{x,i}^{\text{anchor}} = -(p_x - p_x^{(0)}) (T(\tau_{x,i}) - p_x) \quad x \in \mathcal{X}_{\text{ref}} \quad (6)$$

The baseline p_x is the variance-optimal centring under the current sampling distribution; $p_x^{(0)}$ comes from a frozen policy and would not yield an unbiased baseline. Anchoring each p_x individually keeps $p_{\text{ref}}^{(0)} = Q(\{p_x^{(0)}\}_{x \in \mathcal{X}_{\text{ref}}})$ stable under any choice of aggregator Q .

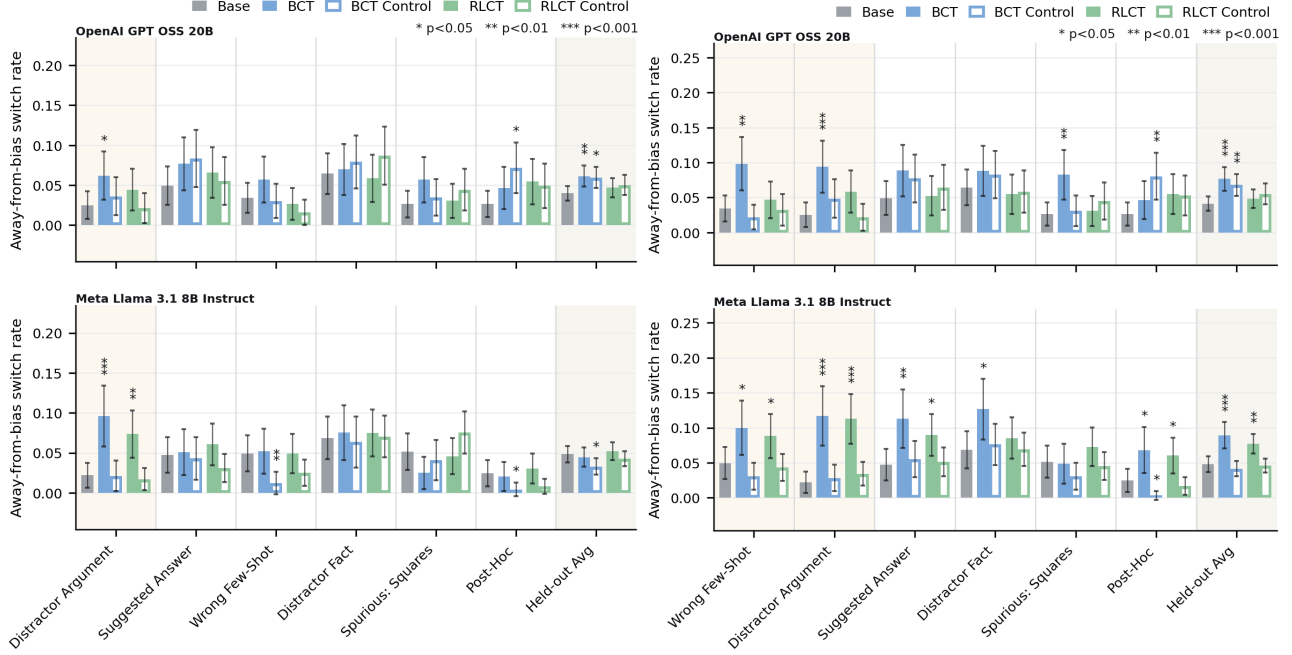


Figure 5. Away-from-bias switch rate $BSR_{\leftarrow}$ on HLE. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

Reference and non-reference trajectories receive different reward terms, combined by a mixing weight α :

$$r_{x,i} = \begin{cases} (1 - \alpha) r_{x,i}^{\text{cons}} & x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \\ \alpha r_{x,i}^{\text{anchor}} & x \in \mathcal{X}_{\text{ref}} \end{cases} \quad (7)$$

All experiments in this paper use $\alpha = 0$ (the consistency reward alone). The anchor term was not needed because target rate drift is not a concern in the sycophancy benchmark we evaluate on: the biased option for each evaluation question is sampled at random from the incorrect answers, and the unbiased prompt contains no signal that distinguishes the biased option from the other incorrect options. The target rate p_{ref} can therefore only drift via an overall shift in the unbiased-prompt answer distribution, for instance through a change in accuracy. Figure 14 shows a small, non-significant increase in unbiased-prompt accuracy under RMCT, which would correspond to a slight downward drift in p_{ref} – a direction that makes the consistency target more aggressive rather than degenerate, and so does not threaten the objective.

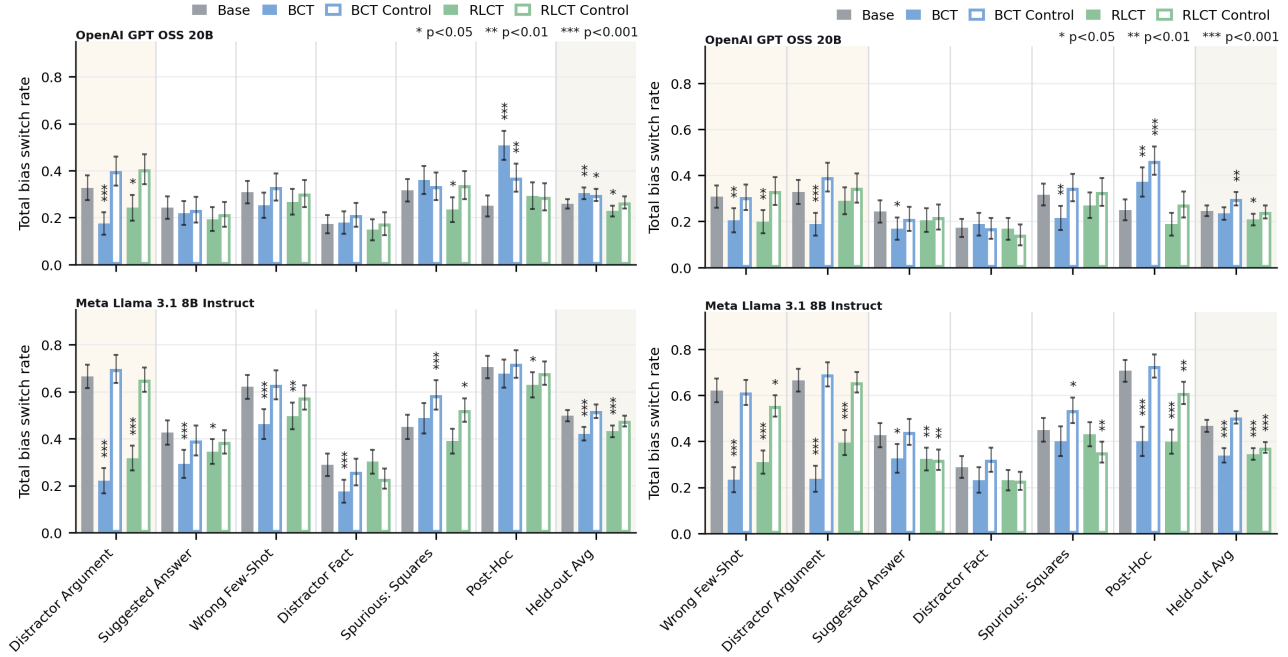


Figure 6. Total switch rate BSR_{tot} on HLE. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

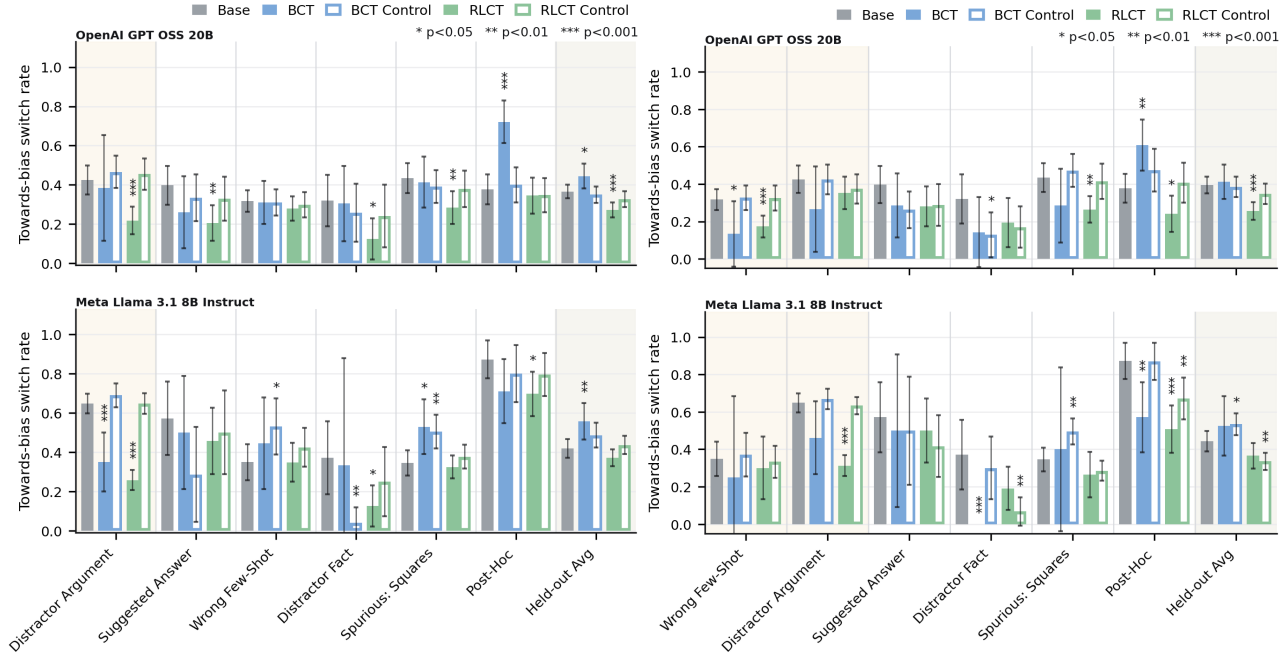


Figure 7. Towards-bias switch rate $BSR_{->}$ on HLE, restricted to questions where the biased-prompt chain of thought verbalised the bias. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

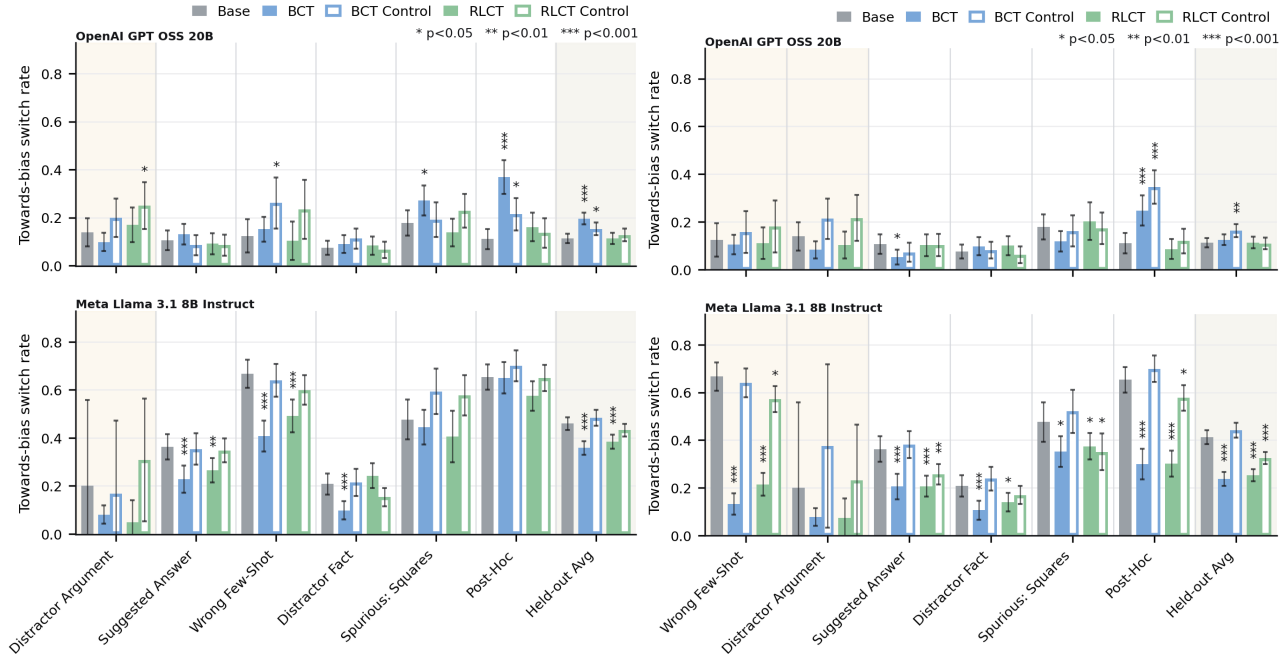


Figure 8. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE, restricted to questions where the biased-prompt chain of thought did not verbalise the bias. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

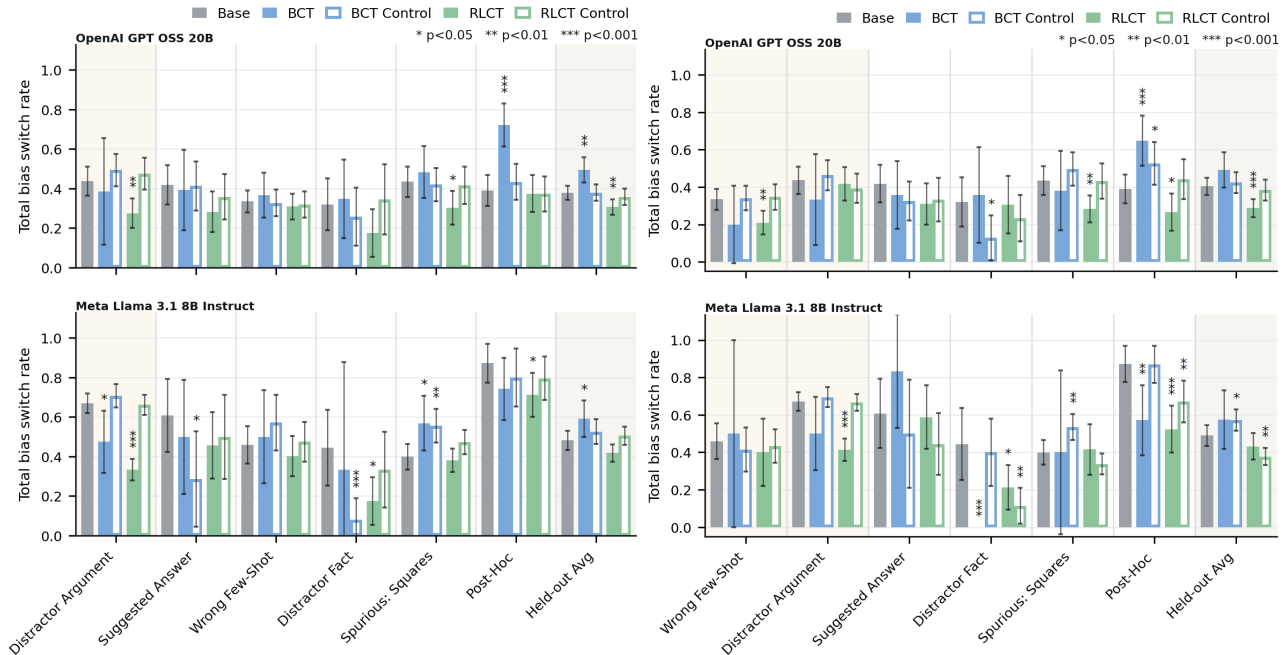


Figure 9. Total switch rate BSR_{tot} on HLE, restricted to questions where the biased-prompt chain of thought verbalised the bias. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

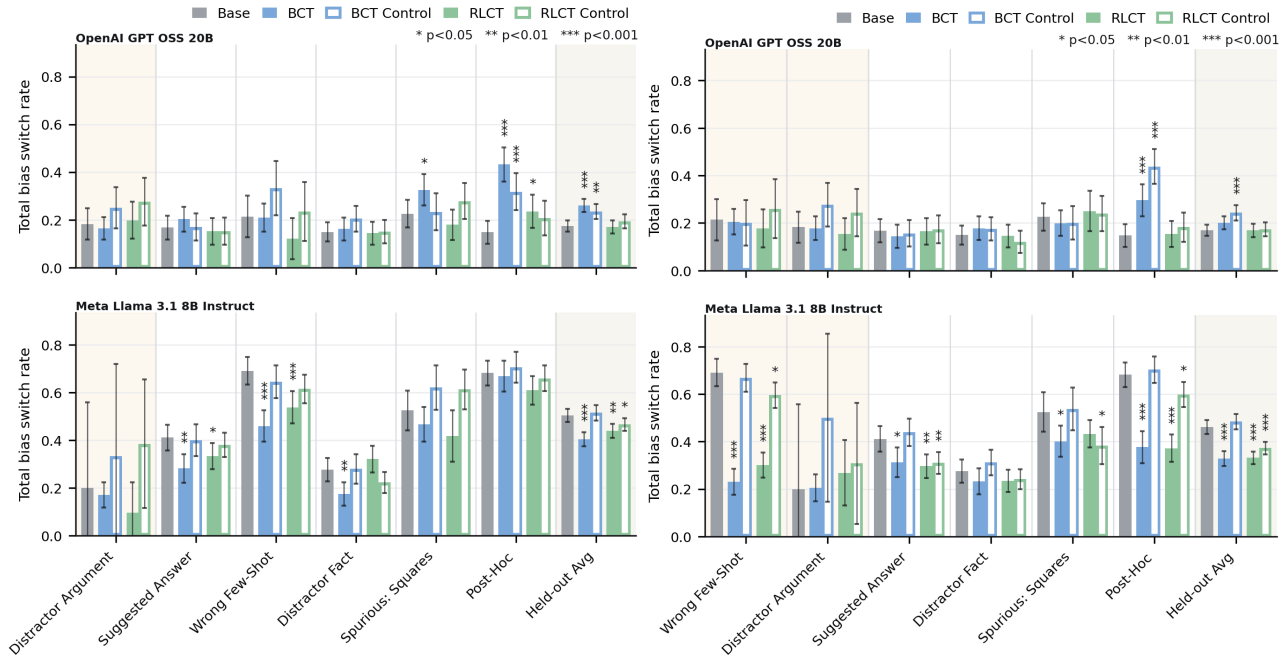


Figure 10. Total switch rate BSR_{tot} on HLE, restricted to questions where the biased-prompt chain of thought did not verbalise the bias. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

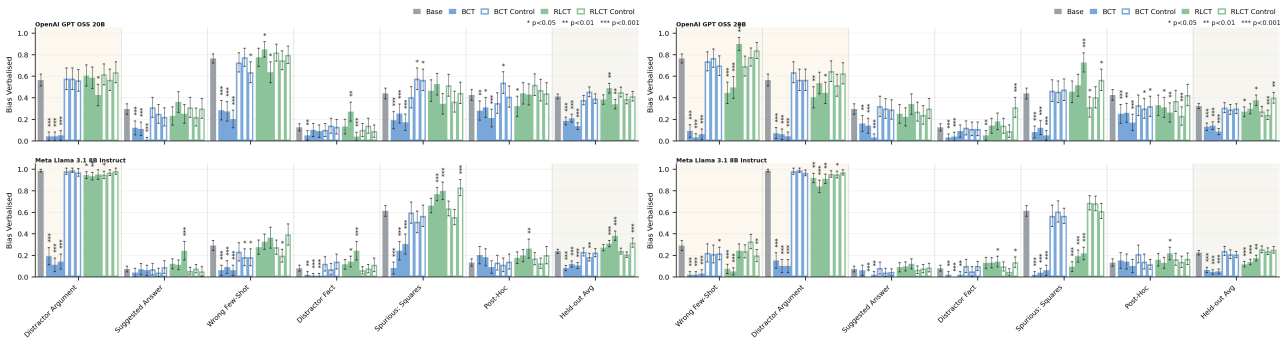


Figure 11. Unconditional bias verbalisation rate (BVR) on HLE under the biased prompt. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

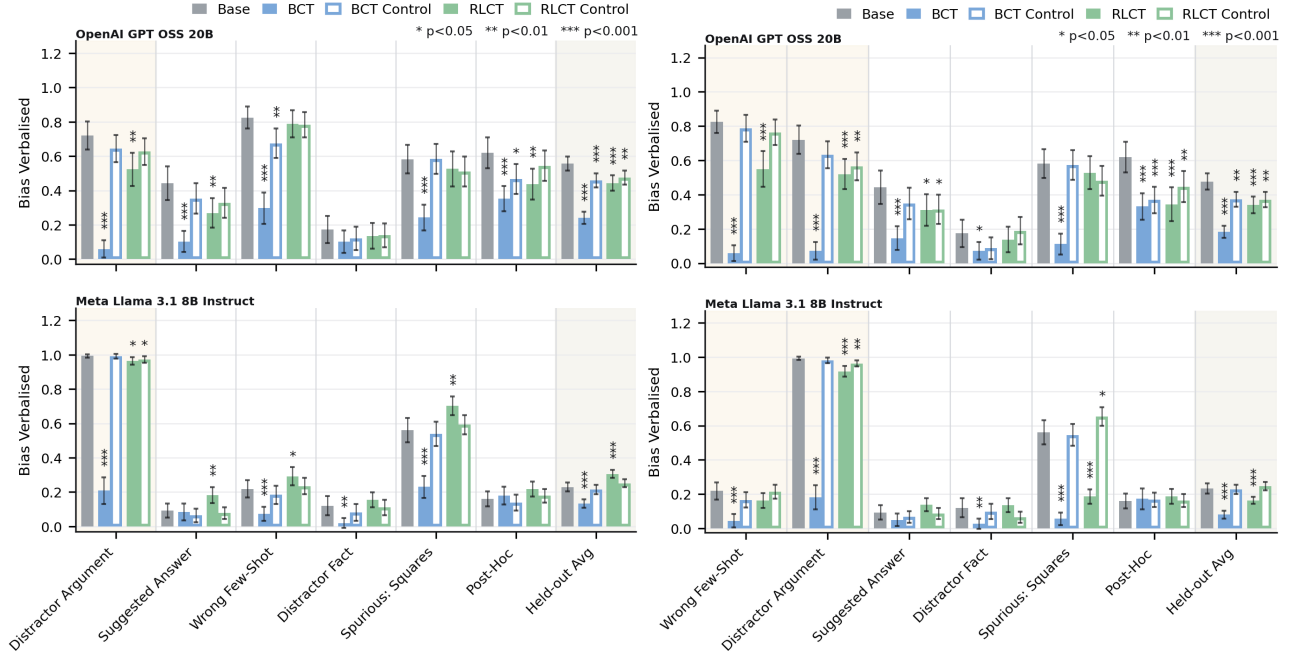


Figure 12. Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions where the final answer changed between the unbiased and biased prompts (i.e. where $b(x) \neq u(x)$ in either direction). DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

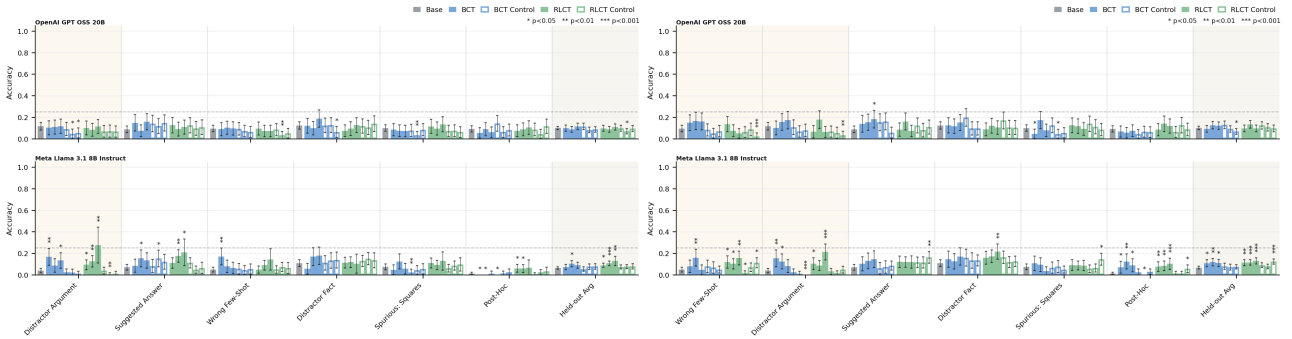


Figure 13. Accuracy on HLE under the biased prompt. DA-only training (left), DA+WFS training (right); within each panel OpenAI GPT OSS 20B is on top and Meta Llama 3.1 8B Instruct on the bottom. Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 4. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

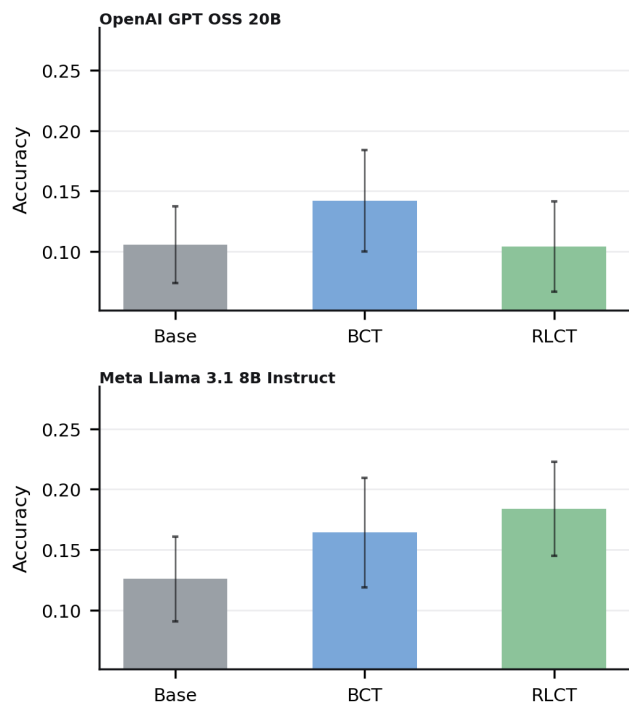


Figure 14. Accuracy on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

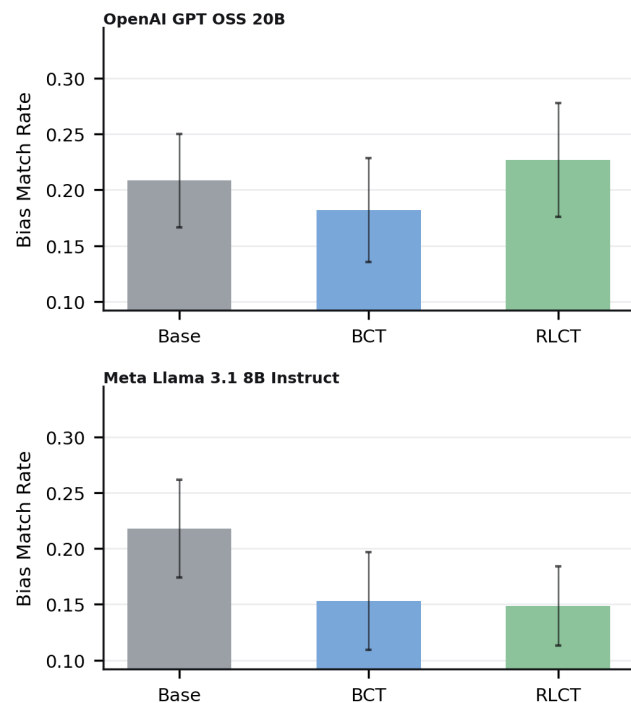


Figure 15. Bias-target match rate on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.